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CUET UG

Applied Mathematics

Financial Mathematics

Perpetuity, Sinking Funds, EMI, CAGR, Depreciation & Bond Valuation · Section B2 Unit VII

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment & Exam Blueprint

In the NTA CUET (UG) Applied Mathematics examination, **Financial Mathematics** (Unit VII of Section B2) forms a heavily weighted, high-scoring section comprising **6 to 8 questions** (30 to 40 marks). Every problem evaluates practical computational fluency, precise conversion of interest periods, and sharp conceptual mastery of time-value-of-money mathematics.

Sub-Topic	Official NTA Syllabus Coverage	High-Yield CUET Focus
Perpetuity & Sinking Funds	Concept of perpetuity and sinking fund; calculating ordinary and growing perpetuity; difference between sinking fund and savings account; sizing periodic sinking fund deposits.	Infinite series summation $\frac{A}{i}$; sinking fund factor $\frac{i}{(1+i)^n-1}$; earmarked legal reserve vs general savings.
EMI Computations	Equated Monthly Installments; reducing-balance method vs flat-rate method; amortization schedule; splitting installment into interest and principal components.	Converting annual rate to monthly period rate ($i = \frac{r}{12}$); loan balance recurrence; effective vs flat interest rate.
Returns, Nominal Rate & CAGR	Calculation of rate of return; nominal rate of return vs effective annual rate; Compound Annual Growth Rate (CAGR); CAGR vs Annual Growth Rate (AGR).	Capital gains plus dividend yield; geometric mean growth $\left(\frac{V_n}{V_0}\right)^{\frac{1}{n}} - 1$; volatility drag in multi-year returns.
Linear Depreciation	Linear (straight-line) depreciation method; cost, residual (scrap) value, and useful life; depreciation rate; book value schedule over asset life.	Depreciable base $(C - S)$; linear annual deduction $\frac{C-S}{n}$; book value after t years $C - tD$.
Bond Valuation	Bond terminology: Face value, coupon rate, coupon payment, maturity, Yield to Maturity (YTM); present value model of coupon bonds and zero-coupon bonds.	Bond pricing formula $V_0 = C[\text{PVIFA}] + F[\text{PVIF}]$; par vs premium vs discount conditions; inverse yield-price relation.

CUET EXAM SCORING STRATEGY — APPLIED FINANCIAL MATHEMATICS

Applied Financial Mathematics questions in CUET reward formula discipline. Over 75% of calculation mistakes arise from two pitfalls: (1) using an annual percentage rate r directly in an EMI or semi-annual bond formula without dividing by compounding frequency ($m = 12$ or $m = 2$), and (2) applying percentages to the wrong base (e.g., computing depreciation on initial cost without subtracting salvage value). Master the period-conversion rules first.

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1. Perpetuity — Valuing Infinite Cash Flows

Definition and Mathematical Formulation

A **perpetuity** is an annuity in which periodic cash flows continue indefinitely ($n \to \infty$). Since the cash flow stream has no terminal date, it has no finite future value; however, because of discounting, it possesses a finite, well-defined **Present Value (PV)**.

Let an ordinary perpetuity provide a periodic payment A at the end of each period, with discount rate i per compounding period. The present value is the sum of an infinite geometric progression:

$$\text{PV} = \frac{A}{1+i} + \frac{A}{(1+i)^2} + \frac{A}{(1+i)^3} + \ldots = \sum_{k=1}^{\infty} \frac{A}{(1+i)^k}$$

Since $\frac{1}{1+i} < 1$ for any positive interest rate $i > 0$, the sum of this infinite series converges to:

$$\text{PV} = \frac{\frac{A}{1+i}}{1 - \frac{1}{1+i}} = \frac{\frac{A}{1+i}}{\frac{i}{1+i}} = \frac{A}{i}$$

PERPETUITY VALUATION FORMULAS

1. **Ordinary Perpetuity (Payments at end of each period):**
$$\text{PV} = \frac{A}{i}$$
2. **Perpetuity Due (Payments at beginning of each period):**
$$\text{PV}_{\text{due}} = A + \frac{A}{i} = A\left(1 + \frac{1}{i}\right) = A\left(\frac{1+i}{i}\right)$$
3. **Growing Perpetuity (Cash flows grow at constant rate $g < i$ per period):**
$$\text{PV}_{\text{growing}} = \frac{A_1}{i - g}$$
where A_1 is the cash flow received at the end of Period 1.

WORKED EXAMPLE 1 — UNIVERSITY CHAIR ENDOWMENT

A philanthropist wishes to establish a perpetual scholarship providing ₹75,000 annually to meritorious students. If the endowment fund earns a steady annual return of 6%, determine:

1. The lump sum required if the first grant is awarded one year from today.
2. The lump sum required if the first grant is awarded immediately today.

Solution:

1. For an ordinary perpetuity:
$$\text{PV} = \frac{A}{i} = \frac{75000}{0.06} = 12,50,000 \text{ rupees.}$$
2. For a perpetuity due (immediate first payout):
$$\text{PV}_{\text{due}} = A + \frac{A}{i} = 75,000 + 12,50,000 = 13,25,000 \text{ rupees.}$$

Deferred Perpetuity

If an ordinary perpetuity is deferred by m periods (meaning the first payment occurs at the end of period $m+1$), its present value evaluated today ($t = 0$) is obtained by discounting the value of the perpetuity at time $t = m$:

$$\text{PV}_{\text{deferred}} = \left(\frac{A}{i}\right) \cdot \frac{1}{(1+i)^m} = \frac{A}{i(1+i)^m}$$

WORKED EXAMPLE 2 — DEFERRED RESEARCH GRANT

A research foundation commits to an endless annual grant of ₹1,20,000, with the first installment payable 4 years from today (i.e. deferred for 3 full years). At an interest rate of 8% compounded annually:

$$\text{PV} = \frac{120000}{0.08} \cdot \frac{1}{(1.08)^3} = 15,00,000 \cdot \frac{1}{1.259712} \approx 11,90,748.83 \text{ rupees.}$$

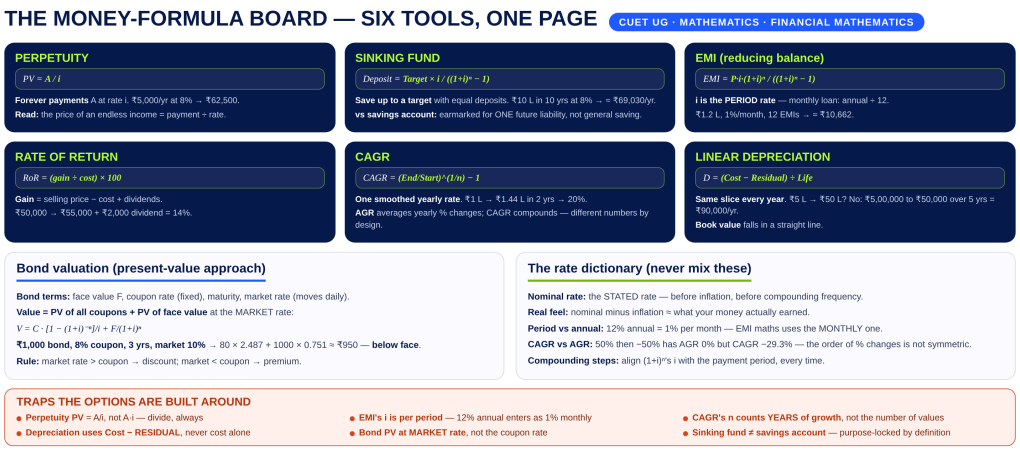


Figure 1: Figure 1: The Financial Mathematics Computational Framework — Perpetuity, Sinking Fund, EMI, Returns, CAGR & Depreciation.

2. Sinking Funds — Systematic Corpus Accumulation

Concept and Core Mechanics

A **sinking fund** is a special-purpose reserve fund created by setting aside periodic, equal payments (R) at compound interest (i) so that the accumulated future value equals a predetermined targeted liability (A) at maturity (n periods).

Sinking funds are mandatory financial instruments in corporate and municipal finance for:

- Redemption of long-term debentures or bonds at maturity.
- Replacement of depreciable fixed capital assets (plant, heavy machinery) at end of useful life.
- Creation of emergency contingency or environmental restoration reserves.

Sinking Fund Mathematical Derivation

Since periodic deposits R represent an ordinary annuity whose future value accumulates to target A:

$$A = R \cdot \frac{(1+i)^n - 1}{i}$$

Solving for the periodic installment R:

$$R = A \cdot \frac{i}{(1+i)^n - 1}$$

The factor $\frac{i}{(1+i)^n-1}$ is termed the **Sinking Fund Factor**.

Comparison Feature	Sinking Fund	Savings Account
Primary Objective	Earmarked contractually for a specific future financial obligation.	General-purpose liquidity, personal savings, or wealth accumulation.
Withdrawal Flexibility	Strictly locked; premature liquidation penalized or legally restricted.	Unrestricted withdrawals permitted at depositor's discretion.
Deposit Schedule	Mathematically computed fixed equal installments targeted at maturity value A.	Arbitrary deposit amounts and frequencies chosen by account holder.
Tax & Accounting Treatment	Often treated as an appropriation of profits / balance sheet reserve.	Personal asset earning taxable interest income.

WORKED EXAMPLE 3 — DEBENTURE REDEMPTION SINKING FUND

InfraCorp Ltd. issues ₹20,00,000 worth of debentures repayable in full after 5 years. A sinking fund earning 10% interest per annum compounded annually is established. Calculate the annual deposit required. (Given: $(1.10)^5 = 1.61051$).

Solution: Target corpus $A = 20,00,000$, $i = 0.10$, $n = 5$.

$$R = 20,00,000 \cdot \frac{0.10}{(1.10)^5 - 1} = 2,00,\frac{000}{1.61051 - 1} = 2,00,\frac{000}{0.61051} \approx 3,27,594.96 \text{ rupees.}$$

3. Equated Monthly Installments (EMI)

Overview and Computational Approaches

An **Equated Monthly Installment (EMI)** is a fixed payment amount made by a borrower to a lender on a specified calendar date each month. EMIs are structured to liquidate both outstanding principal and accrued interest over a specified loan tenure.

There are two primary methods used in financial mathematics to determine EMI:

1. **Flat Rate Method:** Interest is calculated on the initial principal for the full loan term.
2. **Reducing Balance (Amortization) Method:** Interest is computed only on the outstanding principal balance at the beginning of each monthly payment cycle.

The Reducing Balance EMI Formula

Under the reducing balance method, the present value of all monthly installments E must equal the loan principal P:

$$P = E \cdot \frac{1 - (1+i)^{-n}}{i} = E \cdot \frac{(1+i)^n - 1}{i(1+i)^n}$$

Solving explicitly for the monthly installment E:

$$E = P \cdot i \cdot \frac{(1+i)^n}{(1+i)^n - 1}$$

where:

- P = Loan principal borrowed.
- r = Annual interest rate; $i = \frac{r}{12 \cdot 100}$ is the **monthly interest rate**.
- t = Loan tenure in years; $n = 12t$ is the **total number of monthly installments**.

FLAT RATE VS REDUCING BALANCE EMI COMPARISON

- **Flat Rate EMI:**

$$I = P \cdot r \cdot \frac{t}{100}, \quad E_{\text{flat}} = \frac{P+I}{n} = \frac{P\left(1 + r\frac{t}{100}\right)}{12t}$$

- **Reducing Balance EMI:**

$$E_{\text{reducing}} = P \cdot i \cdot \frac{(1+i)^n}{(1+i)^n - 1}$$

Critical CUET Insight: Flat rate loans conceal the true cost of credit. Because the borrower repays principal monthly, the average principal outstanding over the tenure is roughly $\frac{P}{2}$. Consequently, an 8% flat rate corresponds to an effective reducing rate of approximately $14\% - 15\%$.

WORKED EXAMPLE 4 — VEHICLE LOAN EMI

A consumer purchases an electric car by taking a bank loan of ₹6,00,000 at 12% per annum on a reducing balance basis for a tenure of 2 years (24 months). Compute the EMI. (Given: $(1.01)^{24} = 1.26973$).

Solution: Principal $P = 6,00,000$. Monthly interest rate $i = 12\frac{\%}{12} = 1\% = 0.01$. Number of months $n = 24$.

$$E = 6,00,000 \cdot 0.01 \cdot \frac{1.26973}{1.26973 - 1} = 6000 \cdot \frac{1.26973}{0.26973} = \frac{7618.38}{0.26973} \approx 28,244.47 \text{ rupees.}$$

Loan Amortization Schedule Mechanics

In each installment t, the payment E is partitioned into two distinct components:

1. **Interest Component (I_t):** Paid on outstanding opening balance B_{t-1}:
$$I_t = B_{t-1} \cdot i$$
2. **Principal Repayment Component (P_t):** Directly reduces loan principal:
$$P_t = E - I_t$$
3. **Closing Balance (B_t):**
$$B_t = B_{t-1} - P_t$$

As the loan progresses, B_{t-1} decreases monotonically; hence I_t falls steadily, and P_t increases exponentially.

Month (t)	Opening Balance (B_{t-1})	EMI (E)	Interest ($I_t = B_{t-1}i$)	Principal ($P_t = E - I_t$)
1	₹1,00,000.00	₹17,254.80	₹1,000.00	₹16,254.80
2	₹83,745.20	₹17,254.80	₹837.45	₹16,417.35
3	₹67,327.85	₹17,254.80	₹673.28	₹16,581.52
4	₹50,746.33	₹17,254.80	₹507.46	₹16,747.34
5	₹33,998.99	₹17,254.80	₹339.99	₹16,914.81
6	₹17,084.18	₹17,254.80	₹170.84	₹17,083.96

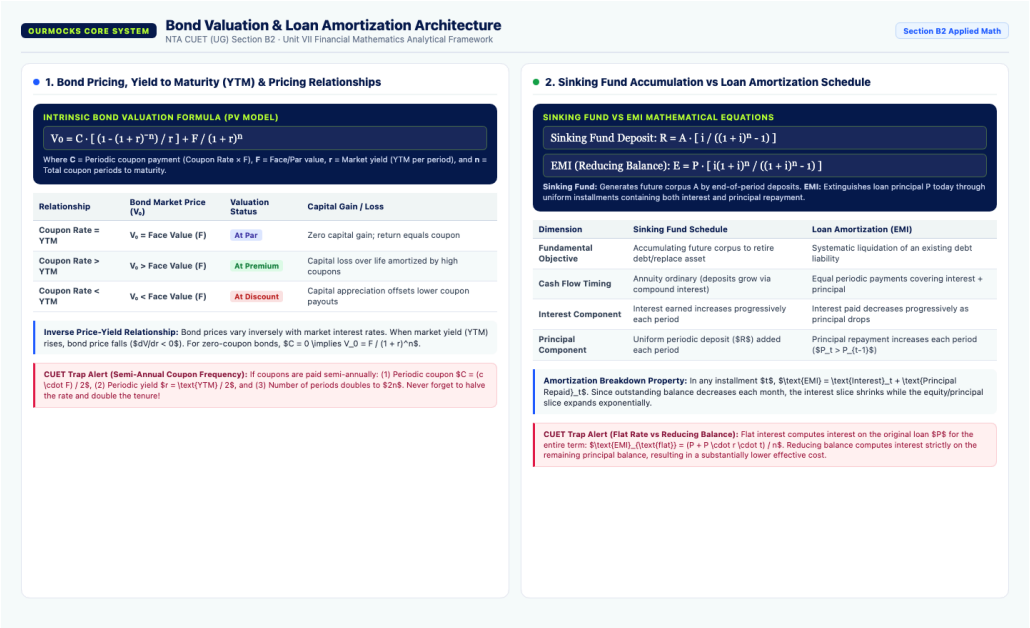


Figure 2: Figure 2: Bond Valuation Dynamics, Price-Yield Relationships & Sinking Fund vs Loan Amortization Schedules.

4. Returns, Nominal vs Effective Rates & CAGR

Rate of Return (RoR)

The **Rate of Return (RoR)** measures the percentage financial gain or loss generated on an investment relative to its initial cost. For any asset holding period:

$$\text{Rate of Return} = \frac{\text{Final Value} - \text{Initial Cost} + \text{Dividends / Cash Income}}{\text{Initial Cost}} \times 100$$

WORKED EXAMPLE 5 — STOCK INVESTMENT RETURN

An investor buys 100 shares of a company at ₹450 per share. Over a 1-year holding period, she receives a total dividend payout of ₹1,800. At the end of the year, she sells all shares at ₹510 per share. Determine her total rate of return.

Solution:

- Initial Investment $= 100 \times 450 = 45,000$ rupees.
- Final Value $= 100 \times 510 = 51,000$ rupees.
- Capital Gain $= 51,000 - 45,000 = 6,000$ rupees.
- Dividend Income $= 1,800$ rupees.
- Total Return $= \frac{6,000+1,800}{45,000} \times 100 = 7,\frac{800}{45},000 \times 100 = 17.33\%$.

Nominal Rate vs Effective Annual Rate (EAR)

When compounding occurs more frequently than once a year (semi-annually, quarterly, monthly), the quoted annual rate is the **nominal rate (r)**. The actual annualized rate earned due to compounding is the **Effective Annual Rate (EAR)**:

$$\text{EAR} = \left(1 + \frac{r}{m}\right)^m - 1$$

where r is the nominal annual rate expressed as a decimal, and m is compounding intervals per year.

Compounding Frequency	Periods (m)	Formula for $1 + \text{EAR}$	EAR for Nominal $r = 12\%$
Annual	1	$(1 + 0.12)^1$	12.000%
Semi-Annual	2	$(1 + 0.06)^2$	12.360%
Quarterly	4	$(1 + 0.03)^4$	12.551%
Monthly	12	$(1 + 0.01)^{12}$	12.683%
Continuous	∞	$e^{0.12}$	12.750%

Compound Annual Growth Rate (CAGR)

The **Compound Annual Growth Rate (CAGR)** provides the constant smoothed annualized geometric growth rate required for an investment to grow from its initial value to its terminal value over n years:

$$\text{Ending Value} = \text{Beginning Value} \cdot (1 + \text{CAGR})^n$$

$$\text{CAGR} = \left(\frac{\text{Ending Value}}{\text{Beginning Value}}\right)^{\frac{1}{n}} - 1$$

CAGR VS ANNUAL GROWTH RATE (AGR) — VOLATILITY DRAG

The **Annual Growth Rate (AGR)** represents the arithmetic mean of individual annual percentage returns:

$$\text{AGR} = \frac{R_1 + R_2 + \ldots + R_n}{n}$$

Because arithmetic averaging ignores compounding, **AGR will always be strictly greater than or equal to CAGR** ($\text{AGR} \geq \text{CAGR}$). The divergence between AGR and CAGR increases with annual return volatility.

- **Example:** If an investment of ₹1,00,000 gains +50% in Year 1 (to ₹1,50,000) and loses −50% in Year 2 (to ₹75,000):

$$\text{AGR} = \frac{+50\% - 50\%}{2} = 0\%$$

$$\text{CAGR} = \left(75, \frac{000}{1}, 00, 000\right)^{\frac{1}{2}} - 1 = \sqrt{0.75} - 1 = 0.8660 - 1 = -13.40\%$$

The investor suffered an actual loss of ₹25,000; CAGR accurately reflects the negative reality, while AGR provides a misleading zero-change illusion.

WORKED EXAMPLE 6 — MUTUAL FUND CAGR CALCULATION

A portfolio valued at ₹2,50,000 on 1st April 2021 grows to ₹5,18,400 on 1st April 2024 (a period of 3 years). Determine the CAGR.

Solution:

$$\text{Beginning Value} = 2, 50, 000, \quad \text{Ending Value} = 5, 18, 400, \quad n = 3.$$

$$\text{Ratio} = 5, 18, \frac{400}{2}, 50, 000 = \frac{5184}{2500} = 2.0736.$$

$$\text{CAGR} = (2.0736)^{\frac{1}{3}} - 1 = (1.2754) - 1 = 27.54\% \text{ per annum.}$$

5. Linear Method of Depreciation

Concepts and Terminology

Depreciation is the systematic allocation of the depreciable cost of a tangible fixed asset over its estimated useful economic life. In Applied Mathematics, the primary technique examined is the **Linear (Straight Line) Method**.

Core parameters:

- **Cost of Asset (C):** Purchase price plus all direct capitalized acquisition costs (freight, installation, import duties, site preparation).
- **Residual / Scrap / Salvage Value (S):** Estimated net realizable value of the asset at the conclusion of its useful life.
- **Useful Life (n):** Estimated operational period (in years) during which the asset remains economically productive.
- **Depreciable Cost / Base:**

$$\text{Depreciable Base} = C - S$$

Depreciation Formulas

LINEAR DEPRECIATION FORMULA SET

1. **Annual Depreciation Amount (D):**

$$D = \frac{C - S}{n}$$

2. **Depreciation Rate (R_D):**

$$R_D = \frac{D}{C} \times 100 = \frac{C - S}{n \cdot C} \times 100\%$$

3. **Accumulated Depreciation after t years (AccD_t):**

$$\text{AccD}_t = t \cdot D = t \cdot \left(\frac{C - S}{n}\right)$$

4. **Book Value after t years (BV_t):**

$$\text{BV}_t = C - \text{AccD}_t = C - t \cdot D$$

At the end of useful life ($t = n$): $\text{BV}_n = C - n\left(\frac{C-S}{n}\right) = S$.

WORKED EXAMPLE 7 — FACTORY MACHINERY DEPRECIATION

A manufacturing unit imports industrial packaging equipment for ₹12,00,000, paying ₹80,000 in transit insurance and ₹1,20,000 for plant commissioning. The equipment has an estimated life of 8 years, after which its scrap value is projected at ₹2,00,000. Calculate:

1. Total capitalized acquisition cost C.
2. Annual depreciation under straight line method.
3. The book value of the machinery at the end of the 5th year.

Solution:

1. $C = 12,00,000 + 80,000 + 1,20,000 = 14,00,000$ rupees.
2. Depreciable Base $= C - S = 14,00,000 - 2,00,000 = 12,00,000$ rupees.

$$D = \frac{12,00,000}{8} = 1,50,000 \text{ rupees per annum.}$$

3. Accumulated Depreciation after 5 years $= 5 \times 1,50,000 = 7,50,000$ rupees.

$$\text{BV}_5 = C - \text{AccD}_5 = 14,00,000 - 7,50,000 = 6,50,000 \text{ rupees.}$$

6. Valuation of Bonds

Bond Terminology & Mechanics

A **bond** is a long-term debt instrument issued by corporations, municipalities, or sovereign governments to raise capital from investors. A standard coupon bond binds the issuer to pay periodic interest (coupons) and redeem the face value at maturity.

Fundamental Parameters:

- **Face Value (F):** Par value stated on the certificate (typically ₹100 or ₹1,000 in Indian bond markets), repaid at maturity.
- **Coupon Rate (c):** Contractual annual nominal interest rate printed on the bond certificate.
- **Coupon Payment (C):** Cash paid per coupon period: $C = c \cdot F$ (or $\frac{c \cdot F}{2}$ for semi-annual payments).
- **Maturity (n):** Number of periods remaining until principal redemption.
- **Yield to Maturity (YTM, r):** Required market discount rate or internal rate of return demanded by investors for bonds of comparable risk.

Intrinsic Bond Valuation Formula (Present Value Approach)

The intrinsic economic value (V_0) of a bond equals the present value of all expected future cash flows discounted at the market yield (r):

$$V_0 = \sum_{t=1}^{n} \frac{C}{(1+r)^t} + \frac{F}{(1+r)^n}$$

Applying the annuity present value factor ($\text{PVIFA}_{r,n}$) and lump sum factor ($\text{PVIF}_{r,n}$):

$$V_0 = C \cdot \left[\frac{1-(1+r)^{-n}}{r}\right] + \frac{F}{(1+r)^n}$$

WORKED EXAMPLE 8 — BOND VALUATION WITH ANNUAL COUPON

A 5-year government bond with a face value of ₹1,000 carries an 8% annual coupon. If the current market yield (YTM) for similar securities is 10%, calculate the intrinsic value of the bond. (Given: $(1.10)^{-5} = 0.62092$).

Solution:

- Coupon payment $C = 8\% \times 1000 = 80$ rupees.
- Face value $F = 1,000$ rupees.
- Discount rate $r = 10\% = 0.10$, $n = 5$.
- Present value of coupons:

$$\text{PV}_{\text{coupons}} = 80 \cdot \left[\frac{1-0.62092}{0.10}\right] = 303.264 \text{ rupees}$$

- Present value of face value:

$$\text{PV}_{\text{face}} = 1000 \cdot (1.10)^{-5} = 620.92 \text{ rupees}$$

- Intrinsic Bond Value:

$$V_0 = \text{PV}_{\text{coupons}} + \text{PV}_{\text{face}} = 303.26 + 620.92 = 924.18 \text{ rupees}$$

Because the market yield (10%) exceeds the coupon rate (8%), the bond trades **at a discount** ($V_0 < 1000$).

Special Bond Classes

1. **Zero-Coupon Bonds (Deep Discount Bonds):** No interim coupon payments ($C = 0$). Sold at a substantial discount to par value and redeemed at par F:

$$V_0 = \frac{F}{(1+r)^n}$$

2. **Perpetual Bonds (Consols):** No maturity date ($n \to \infty$). The issuer promises an indefinite stream of coupon payments:

$$V_0 = \frac{C}{r}$$

Bond Pricing Theorems (The Three Rules of Fixed Income)

Relationship Condition	Bond Market Price (V_0)	Trading Status & Capital Return
Coupon Rate = Market YTM ($c = r$)	$V_0 =$ Face Value (F)	**Trades at Par:** Return derives entirely from coupons; zero capital gain/loss.
Coupon Rate > Market YTM ($c > r$)	$V_0 >$ Face Value (F)	**Trades at Premium:** Price exceeds par; higher coupons offset capital loss at maturity.
Coupon Rate < Market YTM ($c < r$)	$V_0 <$ Face Value (F)	**Trades at Discount:** Price below par; lower coupons compensated by capital gain at maturity.

INVERSE RELATIONSHIP RULE

Bond prices and interest rates share an **inverse relationship**: as market yields (YTM) rise, bond prices fall; when market yields decline, bond prices surge ($d\frac{V_0}{d}r < 0$). Longer-maturity bonds exhibit higher percentage price volatility in response to yield shifts than shorter-maturity bonds.

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Chapter Synthesis & Cognitive Mindmap

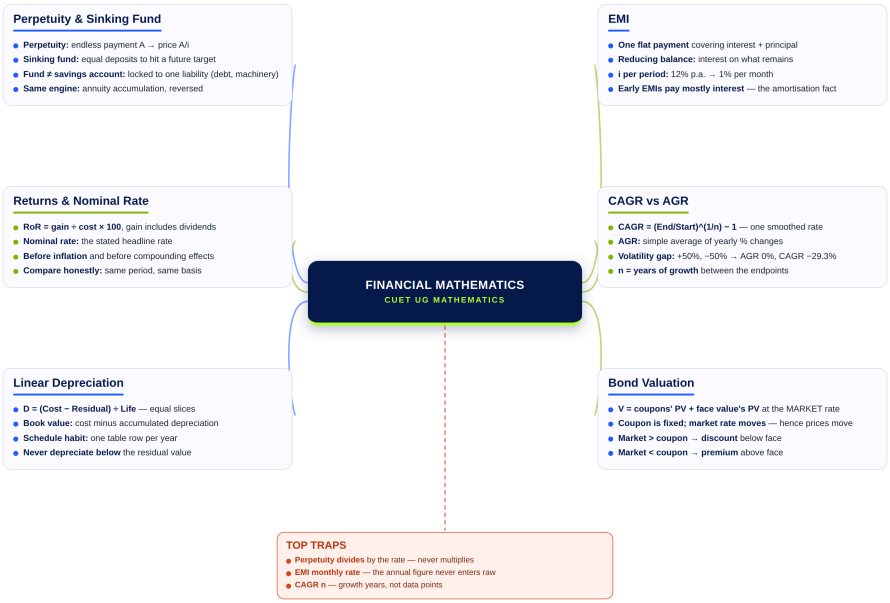


Figure 3: Figure 3: Comprehensive Applied Financial Mathematics Mindmap — Interconnecting All Six Quantitative Pillars.

HIGH-YIELD MEMORY HOOKS FOR CUET 2027

Keep these fundamental mathematical triggers primed:

- **Perpetuity:** Endless stream $\Longrightarrow$ Divide payment by period rate: $\text{PV} = \frac{A}{i}$.
- **Sinking Fund:** Future lump sum $\Longrightarrow$ Multiply target by $\frac{i}{(1+i)^n - 1}$.
- **EMI:** Present loan liquidation $\Longrightarrow$ Multiply principal by $\frac{i(1+i)^n}{(1+i)^n - 1}$.
- **CAGR:** Multi-year growth $\Longrightarrow$ Take n-th root of terminal-to-initial ratio minus 1.
- **SLM Depreciation:** Straight line $\Longrightarrow$ Depreciate net base $(C - S)$ equally over n years.
- **Bonds:** $V_0 = \text{PV of Coupons} + \text{PV of Par}$. If yield exceeds coupon, bond trades at discount.

MASTER FORMULA & DECISION CHEATSHEET — FINANCIAL MATHEMATICS

1. Perpetuity Equations:

Ordinary: $\text{PV} = \dfrac{A}{i}$, Due: $\text{PV} = A\left(\dfrac{1+i}{i}\right)$, Growing: $\text{PV} = \dfrac{A_1}{i-g}$, Deferred: $\text{PV} = \dfrac{A}{i(1+i)^m}$

2. Sinking Fund Annuity:

$$R = A \cdot \frac{i}{(1+i)^n - 1} \qquad (R = \text{Periodic Deposit}, A = \text{Target Future Corpus})$$

3. Loan Amortization & EMI:

$$E_{\text{reducing}} = P \cdot \frac{i(1+i)^n}{(1+i)^n - 1}, \qquad E_{\text{flat}} = \frac{P\left(1 + r\frac{t}{100}\right)}{12t}, \qquad I_t = B_{t-1} \cdot i, \qquad P_t = E - I_t$$

4. Return & Growth Metrics:

$$\text{RoR} = \frac{V_n - V_0 + D}{V_0} \times 100\%, \qquad \text{EAR} = \left(1 + \frac{r}{m}\right)^m - 1, \qquad \text{CAGR} = \left(\frac{V_n}{V_0}\right)^{\frac{1}{n}} - 1$$

5. Straight Line Depreciation:

$$D = \frac{C - S}{n}, \qquad R_D = \frac{D}{C} \times 100\%, \qquad \text{BV}_t = C - t \cdot D$$

6. Bond Valuation Model:

$$V_0 = C \cdot \left[\frac{1 - (1+r)^{-n}}{r}\right] + \frac{F}{(1+r)^n}, \qquad V_0^{\text{zero}} = \frac{F}{(1+r)^n}, \qquad V_0^{\text{perpetual}} = \frac{C}{r}$$

THE SEVEN FATAL CUET TRAPS IN FINANCIAL MATHEMATICS

1. **Compounding Rate Frequency Blindness:** Quoting an annual rate (e.g. 12%) in an EMI calculation without dividing by 12 ($i = 0.01$) or in a semi-annual bond without dividing by 2 ($r = 0.06$).
2. **Salvage Value Omission in Depreciation:** Computing annual depreciation as $\frac{C}{n}$ rather than $\frac{C-S}{n}$. Remember: scrap value is never depreciated!
3. **Depreciation Rate Denominator Confusion:** Calculating depreciation rate as $\frac{D}{C-S}$ instead of $\frac{D}{C}$. The official statutory rate is always expressed as a percentage of initial acquisition cost C.
4. **Flat Rate vs Reducing Balance Deception:** Assuming a flat 10% loan is cheaper than a 14% reducing loan. Because principal is paid down monthly, a 10% flat rate has an effective reducing APR near 18%!
5. **AGR vs CAGR Substitution Error:** Arithmetic averaging of volatile percentage returns (e.g., +20% and −20%) gives 0%, whereas the compounding CAGR is $\sqrt{1.2 \times 0.8} - 1 = \sqrt{0.96} - 1 = -2.02\%$.
6. **Bond Par vs Coupon vs Yield Disconnect:** Forgetting that coupon rate is fixed on face value F, while market price fluctuates based on market yield. When $r > c$, $V_0 < F$ (Discount); when $r < c$, $V_0 > F$ (Premium).
7. **Sinking Fund vs Savings Account Purpose:** Confusing general cash reserves with sinking funds. A sinking fund is contractually designated to retire a specific future debt and cannot be used for operational expenses.

Practice Questions (Exam Pattern)

Fifteen authentic CUET UG examination-standard multiple-choice questions testing conceptual, computational, and institutional mastery of financial mathematics. Marking scheme: +5 for correct answer, −1 for incorrect attempt.

Part 1: Perpetuity, Sinking Funds & Loan Amortization

Q1. A philanthropic trust intends to institute an annual prize of ₹18,000 in perpetuity to the topper of an engineering college. If the corpus fund is invested in government securities yielding an annual return of 7.5%, the minimum lump sum donation required today is:

(A) ₹2,25,000 **(B)** ₹2,40,000
(C) ₹1,35,000 **(D)** ₹2,50,000

Sol. Q1. For an ordinary perpetuity, the present value of endless periodic payouts A discounted at annual rate i is:

$$\text{PV} = \frac{A}{i}$$

. Here, $A = 18,000$ and $i = 7.5\% = 0.075 = \frac{3}{40}$. Thus,

$$\text{PV} = \frac{18000}{\frac{3}{40}} = 18000 \times \frac{40}{3} = 2,40,000 \text{ rupees}$$

. **Answer: (B)**

Q2. A company needs to replace a transport fleet in 4 years at an estimated cost of ₹15,00,000. It establishes a sinking fund earning 8% interest per annum compounded annually. The equal annual deposit required at the end of each year is: (Given: $(1.08)^4 = 1.3605$)

(A) ₹3,75,000.00 **(B)** ₹4,08,150.00
(C) ₹3,32,871.01 **(D)** ₹3,00,000.00

Sol. Q2. Target future corpus $A = 15,00,000$, interest rate $i = 0.08$, tenure $n = 4$ years. The periodic sinking fund deposit R is:

$$R = A \cdot \frac{i}{(1+i)^n - 1} = 15,00,000 \cdot \frac{0.08}{1.3605 - 1} = 1,20,\frac{000}{0.3605} \approx 3,32,871.01 \text{ rupees}$$

. Matches Option C. **Answer: (C)**

Q3. A housing finance company offers a home improvement loan of ₹3,00,000 repayable in 36 equated monthly installments at 12% per annum on a reducing balance basis. The monthly interest rate (i) and the number of periods (n) to be utilized in the EMI formula are:

(A) $i = 0.12, \quad n = 3$ **(B)** $i = 0.06, \quad n = 6$
(C) $i = 0.001, \quad n = 36$ **(D)** $i = 0.01, \quad n = 36$

Sol. Q3. In monthly installment loan calculations: $i = \frac{r}{12 \cdot 100} = \frac{12}{1200} = 0.01$ (or 1% per month) and $n =$ Tenure in years $\times 12 = 3 \times 12 = 36$ months. Matches Option D precisely. Using $i = 0.12$ directly is a classic trap. **Answer: (D)**

Q4. The primary legal and structural difference between a sinking fund and an ordinary savings bank account is that a sinking fund:

(A) Pays no interest on accumulated balances **(B)** Is legally earmarked for the systematic liquidation of a specific predetermined future obligation
(C) Allows unlimited daily debit withdrawals via automated teller machines **(D)** Is exclusively available to non-profit charitable societies

Sol. Q4. A sinking fund is a specialized financial device whose sole objective is accumulating capital to retire a specific designated liability (e.g., bond redemption or plant replacement) at a fixed maturity date. Withdrawals are strictly restricted to that designated purpose. **Answer: (B)**

Q5. Under the reducing balance method of loan amortization, which of the following statements correctly characterizes the behavior of the interest and principal components across the loan tenure?

(A) Both interest and principal components remain strictly constant in every installment. **(B)** The interest component increases progressively while the principal component decreases.
(C) The interest component decreases progressively while the principal component increases. **(D)** The total installment amount (E) fluctuates depending on monthly bank base rate revisions.

Sol. Q5. In an amortization schedule with fixed EMI $E = I_t + P_t$, the interest component $I_t = B_{t-1} \cdot i$ decreases monotonically as principal is repaid. To maintain a constant EMI E, the principal repayment component $P_t = E - I_t$ must progressively increase. **Answer: (C)**

Part 2: Returns, Rates, CAGR & Depreciation

Q6. An investor purchased 200 equity shares of a listed pharmaceutical firm at ₹350 per share. Over a 1-year holding period, she collected ₹15 per share in interim cash dividends. At the end of the year, she liquidated her entire holding at ₹406 per share. Her total holding period rate of return is:

(A) 16.0% (B) 18.5%
(C) 20.3% (D) 22.1%

Sol. Q6. Total Initial Cost $= 200 \times 350 = 70,000$ rupees. Final Value $= 200 \times 406 = 81,200$ rupees. Capital Gain $= 81,200 - 70,000 = 11,200$ rupees. Dividend Income $= 200 \times 15 = 3,000$ rupees. Total Return $= \left(14, \frac{200}{70}, 000\right) \times 100 = 20.2857\% \approx 20.3\%$. Matches Option C. **Answer: (C)**

Q7. A venture capital investment in an enterprise grew from an initial valuation of ₹4,00,000 to ₹13,50,000 over a period of 3 years. The Compound Annual Growth Rate (CAGR) of this venture is: (Given: $(3.375)^{\frac{1}{3}} = 1.50$)

(A) 79.2% (B) 42.5%
(C) 33.3% (D) 50.0%

Sol. Q7. Initial value $V_0 = 4,00,000$, final value $V_3 = 13,50,000$, tenure $n = 3$ years.

$$\text{CAGR} = \left(\frac{V_3}{V_0}\right)^{\frac{1}{n}} - 1 = \left(13, 50, \frac{000}{4}, 00, 000\right)^{\frac{1}{3}} - 1 = (3.375)^{\frac{1}{3}} - 1 = 1.50 - 1 = 0.50 = 50.0\%$$

. Matches Option D. **Answer: (D)**

Q8. A high-growth equity fund registered a return of +40% in Year 1, followed by a sharp contraction of −30% in Year 2. The Annual Growth Rate (AGR, arithmetic mean) and the Compound Annual Growth Rate (CAGR) for the 2-year period are:

(A) AGR = 5.0%, CAGR ≈ −1.0% (B) AGR = 10.0%, CAGR = 5.0%
(C) AGR = 5.0%, CAGR = 5.0% (D) AGR = −1.0%, CAGR = 5.0%

Sol. Q8.

1. Annual Growth Rate (Arithmetic Mean):

$$\text{AGR} = \frac{+40\% - 30\%}{2} = 5.0\%$$

. 2. Compound Annual Growth Rate: If initial value is 100, Year 1 value is 140, Year 2 value is $140 \times 0.70 = 98$.

$$\text{CAGR} = \sqrt{\frac{98}{100}} - 1 = \sqrt{0.98} - 1 \approx 0.98995 - 1 \approx -1.0\%$$

.

Answer: (A)

Q9. A high-precision CNC machine was procured for ₹10,00,000. Its estimated useful economic life is 6 years, with an anticipated scrap value of ₹1,00,000. Under the linear (straight line) method, the annual depreciation and the book value at the end of the 4th year are:

(A) Annual Depreciation = ₹1,50,000; Book Value = ₹4,00,000 (B) Annual Depreciation = ₹1,66,667; Book Value = ₹3,33,333
(C) Annual Depreciation = ₹1,50,000; Book Value = ₹4,50,000 (D) Annual Depreciation = ₹1,80,000; Book Value = ₹2,80,000

Sol. Q9. Acquisition cost $C = 10,00,000$, Scrap value $S = 1,00,000$, Useful life $n = 6$ years. Depreciable Base $= C - S = 9,00,000$. Annual Depreciation $D = 9,00, \frac{000}{6} = 1,50,000$. Accumulated Depreciation after 4 years $= 4 \times 1,50,000 = 6,00,000$. Book Value $= 10,00,000 - 6,00,000 = 4,00,000$. **Answer: (A)**

Q10. A corporate bond carrying a coupon rate of 9% payable annually is currently quoted in the debt market. If the prevailing market Yield to Maturity (YTM) for comparable risk-class debt instruments is 7.5%, the bond will trade:

(A) At Par ($V_0 = F$) (B) At a Discount ($V_0 < F$)
(C) At a Premium ($V_0 > F$) (D) At Zero Market Value

Sol. Q10. When contractual coupon rate ($c = 9\%$) exceeds market yield ($r = 7.5\%$), the bond delivers above-market cash flows. Investors bid up its price above face value ($V_0 > F$). Therefore, the bond trades at a premium. Matches Option C. **Answer: (C)**

Part 3: Bond Valuation & Fixed Income Mathematics

Q11. A zero-coupon bond has a face value of ₹10,000 and matures after 3 years. If the prevailing market interest rate is 8% compounded annually, the fair present market value of the bond is: (Given: $(1.08)^{-3} = 0.79383$)

(A) ₹7,938.30 **(B)** ₹8,000.00
(C) ₹9,200.00 **(D)** ₹6,830.13

Sol. Q11. For a zero-coupon bond, $C = 0$. The valuation formula is the present value of face value:

$$V_0 = \frac{F}{(1+r)^n} = F \cdot (1+r)^{-n}$$

. For $F = 10,00,000$, $r = 0.08$, $n = 3$:

$$V_0 = 10,000 \times 0.79383 = 7,938.30 \text{ rupees}$$

. **Answer: (A)**

Q12. A perpetual government bond pays an annual coupon of ₹600 indefinitely. If the required market yield increases from 5% to 6%, the percentage change in the market price of the consol is:

(A) Increases by 20.0% **(B)** Decreases by 10.0%
(C) Decreases by 20.0% **(D)** Decreases by 16.67%

Sol. Q12. The price of a perpetual bond is $V_0 = \frac{C}{r}$. At $r_1 = 5\% = 0.05$: $V_1 = \frac{600}{0.05} = 12,000$. At $r_2 = 6\% = 0.06$: $V_2 = \frac{600}{0.06} = 10,000$. Change in price $= 10,000 - 12,000 = -2,000$. Percentage change $= \left(-2, \frac{000}{12}, 000\right) \times 100 = -16.67\%$. Matches Option D. **Answer: (D)**

Q13. A machine costing ₹5,00,000 is depreciated at 10% per annum using the straight-line method. If the scrap value at the end of its useful life is ₹50,000, the estimated useful life of the machine is:

(A) 9 years **(B)** 10 years
(C) 8 years **(D)** 12 years

Sol. Q13. In linear depreciation, annual depreciation $D = C \times \frac{R_D}{100} = 5,00,000 \times 0.10 = 50,000$ rupees per annum. Total depreciable base $= C - S = 5,00,000 - 50,000 = 4,50,000$. Useful life $n = \frac{C-S}{D} = 4,50, \frac{000}{50}, 000 = 9$ years. **Answer: (A)**

Q14. A commercial bank offers a nominal deposit rate of 12% per annum. Which compounding frequency yields an Effective Annual Rate (EAR) closest to 12.68%? (Given: $(1.03)^4 = 1.12551$, $(1.01)^{12} = 1.12683$)

(A) Compounded Annually **(B)** Compounded Monthly
(C) Compounded Semi-Annually **(D)** Compounded Quarterly

Sol. Q14. The Effective Annual Rate is EAR $= \left(1 + \frac{r}{m}\right)^m - 1$. For quarterly compounding ($m = 4$): EAR $= (1.03)^4 - 1 = 12.551\%$. For monthly compounding ($m = 12$): EAR $= (1.01)^{12} - 1 = 12.683\% \approx 12.68\%$. Matches Option B. **Answer: (B)**

Q15. In an equated monthly installment (EMI) loan of ₹1,00,000 taken at 12% per annum reducing balance for 12 months with an EMI of ₹8,884.88, the interest portion and the principal portion of the very first monthly installment are:

(A) Interest = ₹12,000.00; Principal = −₹3,115.12 **(B)** Interest = ₹1,000.00; Principal = ₹7,884.88
(C) Interest = ₹888.49; Principal = ₹7,996.39 **(D)** Interest = ₹4,442.44; Principal = ₹4,442.44

Sol. Q15. Opening balance $B_0 = 1,00,000$. Monthly interest rate $i = 12\frac{\%}{12} = 1\% = 0.01$. Interest in month 1: $I_1 = B_0 \cdot i = 1,00,000 \times 0.01 = 1,000$. Principal repaid in month 1: $P_1 = E - I_1 = 8,884.88 - 1,000 = 7,884.88$. Matches Option B. **Answer: (B)**

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